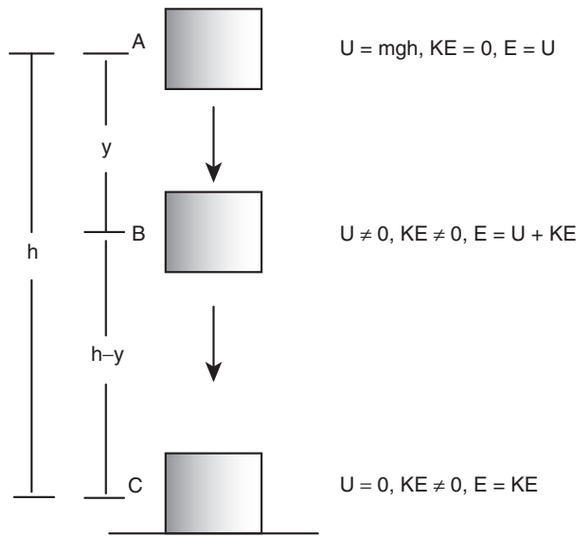


Figure 4.13 Conservation of energy

The law of conservation of energy states that *energy can neither be created nor destroyed. It may be transformed from one form to another but the total energy of an isolated system remains constant.*

Figure 4.13 illustrates that, if an object starts from rest at height *h*, the total energy is purely potential energy (U=mgh) and the kinetic energy (KE) is zero at *h*. When the object falls at some distance *y*, the potential energy and the kinetic energy are not zero whereas, the total energy remains same as measured at height *h*. When the object is about to touch the ground, the potential energy is zero and total energy is purely kinetic.

EXAMPLE 4.13

An object of mass 1 kg is falling from the height h = 10 m. Calculate

(a) The total energy of an object at $h = 10$ m
(b) Potential energy of the object when it is at $h = 4$ m
(c) Kinetic energy of the object when it is at $h = 4$ m
(d) What will be the speed of the object when it hits the ground?

(Assume $g = 10\ \text{m s}^{-2}$)

Solution

(a) The gravitational force is a conservative force. So the total energy remains constant throughout the motion. At $h = 10$ m, the total energy E is entirely potential energy.

$$E = U = mgh = 1 \times 10 \times 10 = 100\ \text{J}$$

(b) The potential energy of the object at $h = 4$ m is

$$U = mgh = 1 \times 10 \times 4 = 40\,\text{J}$$

(c) Since the total energy is constant throughout the motion, the kinetic energy at $h = 4$ m must be $KE = E - U = 100 - 40 = 60\,\text{J}$

Alternatively, the kinetic energy could also be found from velocity of the object at 4 m. At the height 4 m, the object has fallen through a height of 6 m.

The velocity after falling 6 m is calculated from the equation of motion,

$$v = \sqrt{2gh} = \sqrt{2 \times 10 \times 6} = \sqrt{120}\ \text{m s}^{-1};$$
$$v^2 = 120$$

The kinetic energy is $\text{KE} = \frac{1}{2}mv^2 = \frac{1}{2} \times 1 \times 120 = 60\,\text{J}$

(d) When the object is just about to hit the ground, the total energy is completely kinetic and the potential energy, $U = 0$.

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